

AtliQ Mart

Supply Chain Service-Level Dashboard

End-to-end supply-chain analytics: **6** CSV tables → Power Query star schema → **19** DAX measures → **5**-page Power BI dashboard — with verified numbers and root-cause insights.

31,729

orders

57,096

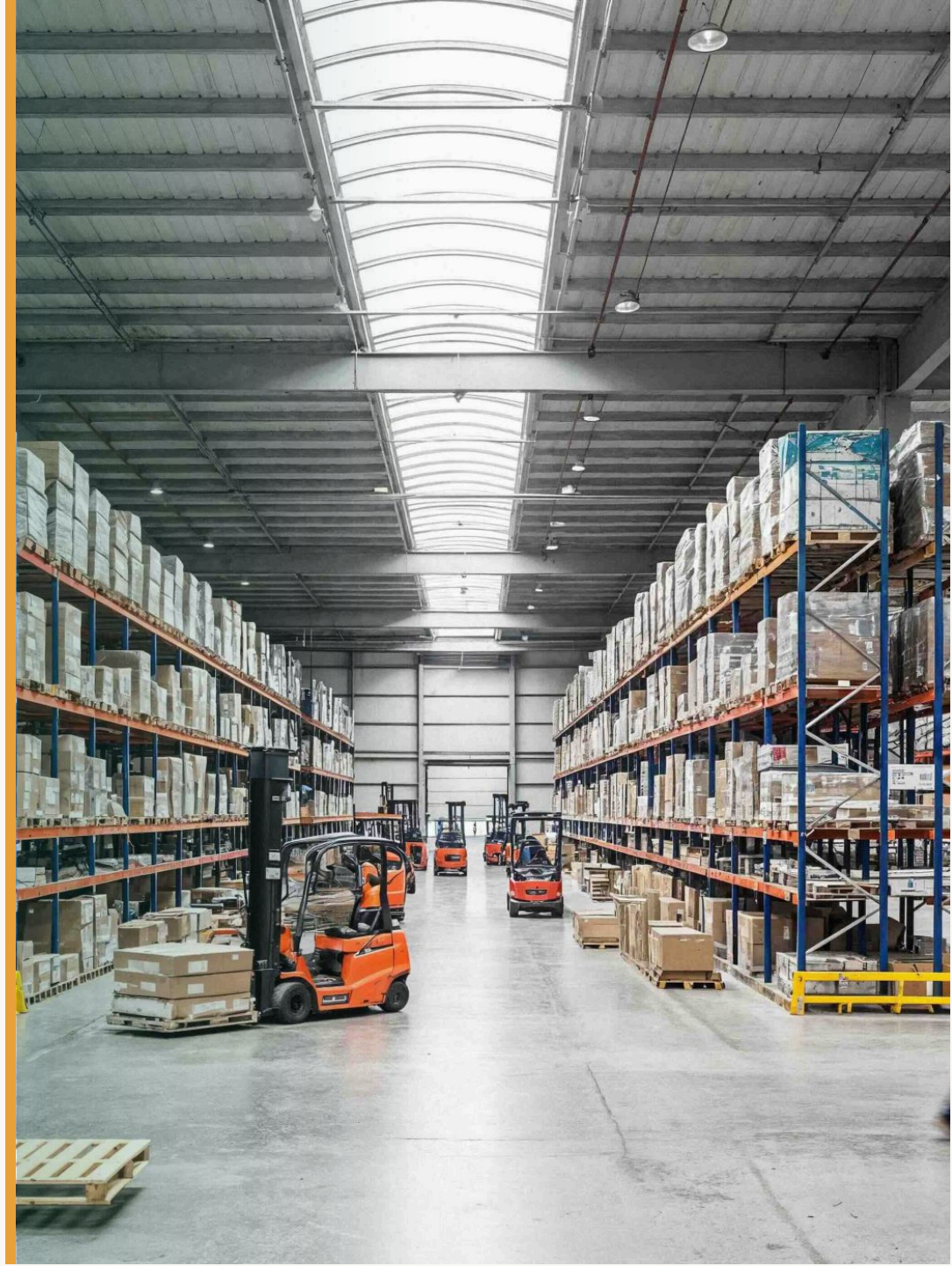
order lines

OTIF **29.0%**

vs 65.9% target

0 / 35

customers on target



THE PROBLEM

Key customers are walking away over late, incomplete deliveries

AtliQ Mart is an FMCG manufacturer in Gujarat, India, supplying retail chains across Surat, Ahmedabad and Vadodara — and about to expand into new metros and Tier-1 cities.

Several key customers have declined to renew their annual supply contracts, citing late and incomplete deliveries. Leadership needed a daily service-level tracking dashboard before the expansion replicates the problem in new cities.

“

Where is order fulfillment failing — for whom, how badly, and why — so the supply chain can be fixed before the expansion replicates the problem in new cities?

WHAT THE DASHBOARD TRACKS

OT%

On-Time

IF%

In-Full

OTIF%

On-Time In-Full

LIFR

Line Fill Rate

VOFR

Volume Fill Rate

- All measured per customer against negotiated targets.



Six CSV tables, six months of order flow

01 Mar – 31 Aug 2022

DATA WINDOW

3

CITIES

Surat · Ahmedabad · Vadodara

35

CUSTOMERS

18

PRODUCTS

in 3 categories

TABLE	ROWS	ROLE
fact_order_lines	57,096	FACT — one row per order line
fact_orders_aggregate	31,729	FACT — one row per order (OT / IF / OTIF flags)
dim_customers	35	DIMENSION — retail customers (chain + city)
dim_products	18	DIMENSION — products and categories
dim_date	183	DIMENSION — daily calendar, Mar–Aug 2022
dim_targets_orders	35	DIMENSION — per-customer OT / IF / OTIF targets



Every figure was recomputed from the raw CSVs during the build — chart exports checked by hand. Headline LIFR 65.96% and VOFR 96.59% reproduce the values widely cited for this dataset, validating the metric definitions.

Five quiet data traps, each fixed at the source

PROBLEM

FIX

01

Invisible BOM characters hiding in `dim_date.date` and `order_id` headers



Headers renamed character-by-character — the invisible bytes would silently break relationships later

02

Two different date formats: `"Tuesday, March 1, 2022"` vs `"01-Mar-22"`



Both parsed to a true Date type, so time-intelligence and the calendar connection work

03

Category casing broken: `"beverages"` vs `"Dairy"` / `"Food"`



Transform → Format → Capitalize Each Word — after Replace Values failed silently by acting on the wrong column

04

% signs polluting target column names
(`ontime_target%`)



Renamed — values already numeric (87 = 87%), the % was decoration that made DAX formulas ugly

05

Spaces in `"In Full"` / `"On Time"` / `"On Time In Full"` columns



Renamed to `in_full` / `on_time` / `on_time_in_full`

A clean star schema — six one-way streets, no fact-to-fact joins

MODEL MAP



- 6 relationships
- all single-direction
- filters flow dimension → fact only

KEY DECISIONS



Deleted the auto-created fact-to-fact order_id relationship

Facts must never filter each other; two competing filter paths produce ambiguous numbers



Rebuilt dim_customers ↔ dim_targets_orders as one-to-many, single direction

The arrow must point customers → targets, or every matrix row shows the same grand-average target



mmm_yy sorted alphabetically (Apr before Feb)

Added a month_sort column in Power Query and set it as the sort-by column



Disabled Auto date/time

The model uses its own dim_date — no hidden calendar tables

19 explicit measures — zero implicit aggregations

```
THE CORE KPI
OTIF % =
DIVIDE (
    SUM ( fact_orders_aggregate[otif] ),
    [Total Orders]
)

GAP IN POINTS
OTIF Gap =
( [OTIF %] - [OTIF Target %] ) * 100

TRAFFIC-LIGHT COLOR, WIRED INTO MATRIX CELLS
OTIF Gap Color =
VAR GapPoints = [OTIF Gap]
RETURN SWITCH ( TRUE (),
    GapPoints >= 0, "#63BE7B",
    GapPoints >= -10, "#FFEB84",
    GapPoints >= -25, "#FDAE61",
    "#F8696B"
)
```

🔗 A disconnected Metrics table + SELECTEDVALUE router powers the metric switcher on the trend charts.

Σ Measure inventory 19 total

2

base totals

5

service-level KPIs

3

targets

3

gaps (pts)

3

conditional colors

2

switcher routers + dynamic chart title

- <> DAX HABITS
- ✓

DIVIDE over / — safe divide-by-zero
- ✓

SUM on 0/1 flags as counters
- ✓

SWITCH (TRUE()) ladders
- ✓

VAR ... RETURN
- ✓

SELECTEDVALUE + disconnected table

Five pages, one story — starting with the one-screen answer

01

Overview

executive readout: five KPIs vs targets, city split, monthly OTIF trend

VIEWING

02

Customer Performance

account matrix + watchlist

03

Trends

metric switcher + month → day drill

04

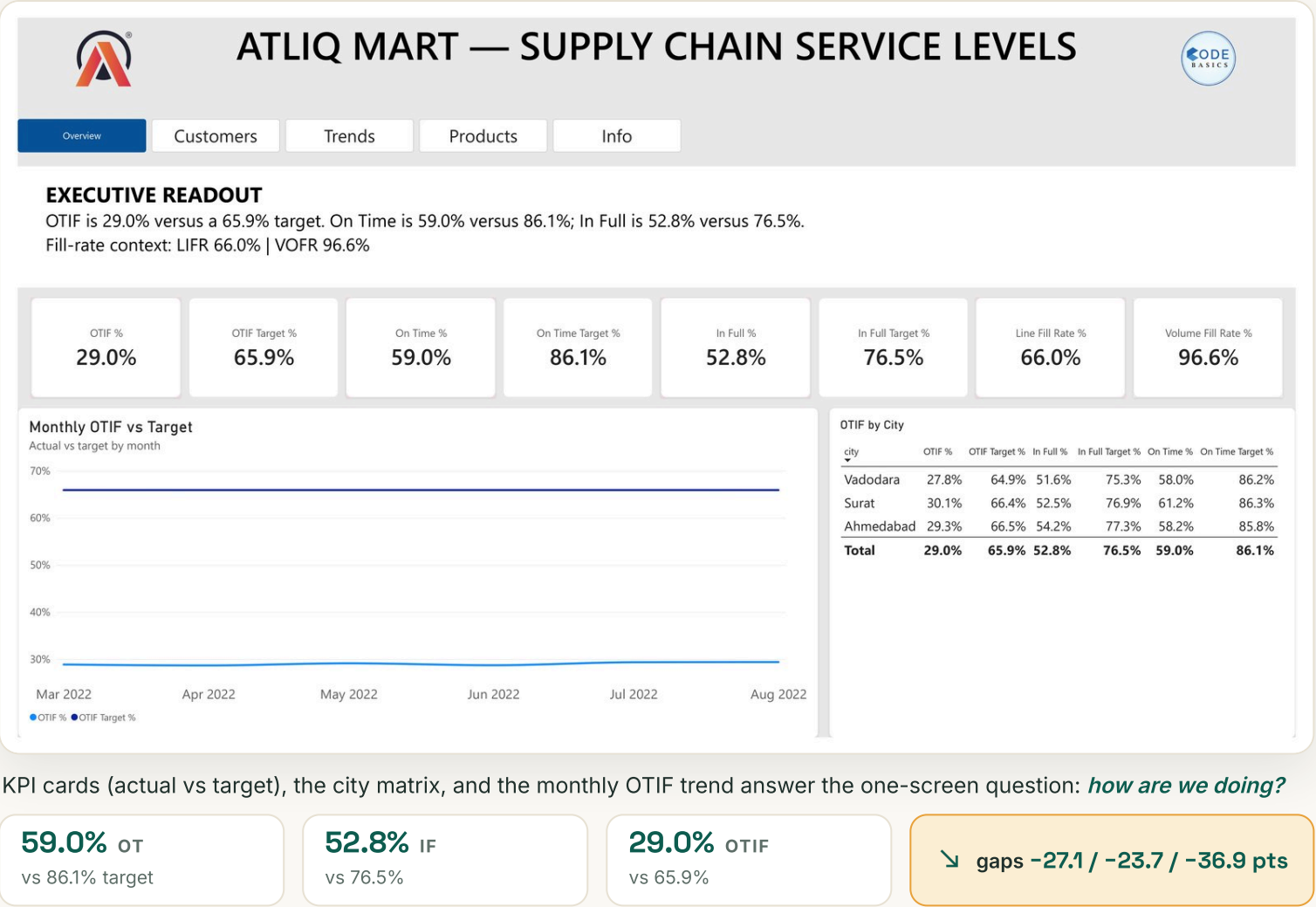
Product Insights

fill rates + sparklines

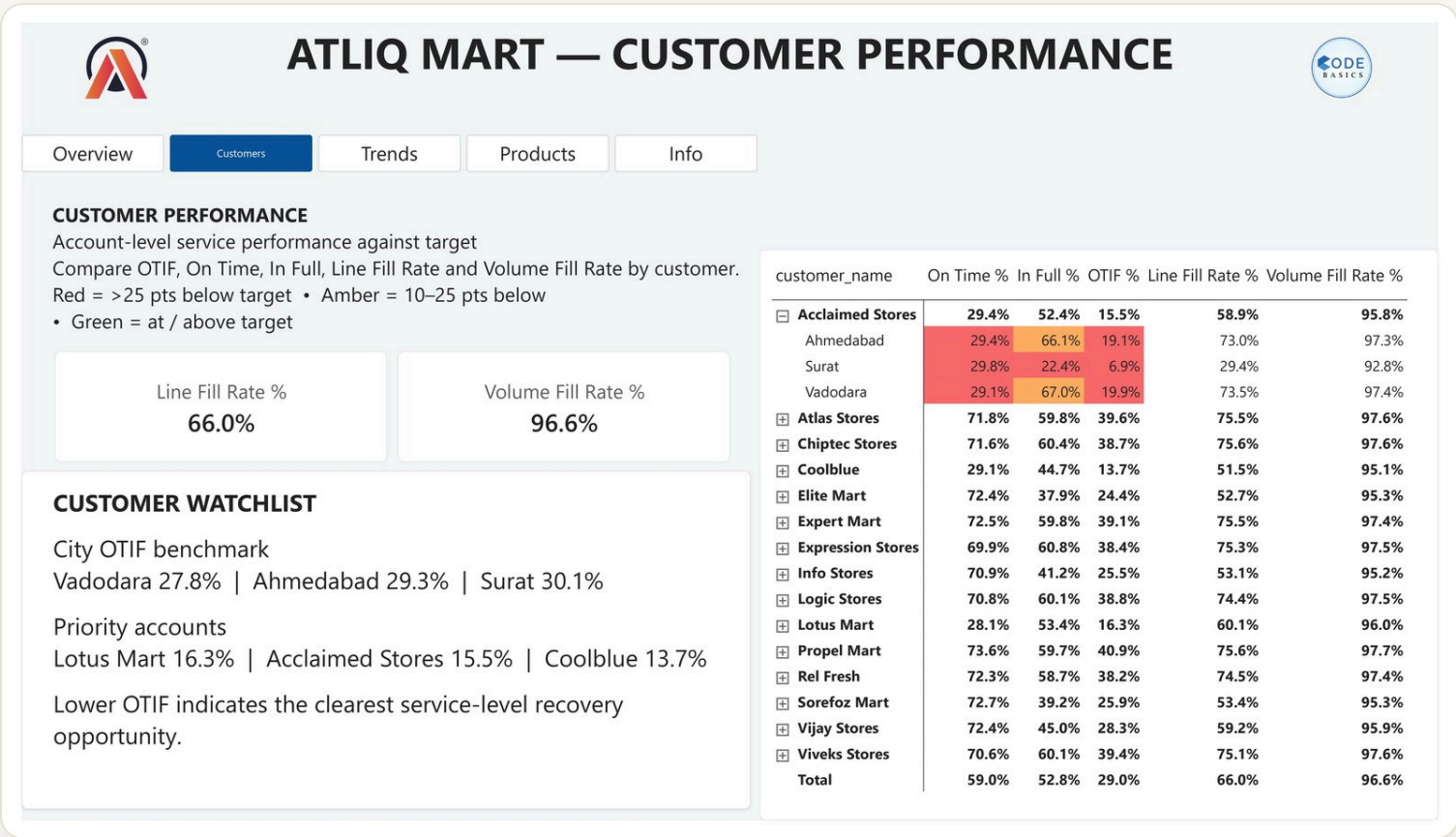
05

Definitions & Data Model

definitions, schema, insights



Every account, every metric, color-coded by gap to target



Account-level matrix

OT / IF / OTIF / LIFR / VOFR for each of the 35 customers, against that customer's negotiated target



Traffic-light cells

Conditional formatting driven by field-value measures

- #63BE7B

at/above target
- #FFEB84

within 10 pts
- #FDAE61

within 25 pts
- #F8696B

beyond 25 pts



Customer Watchlist

City OTIF benchmark plus the lowest-OTIF priority accounts — the renewal-risk shortlist



The verdict: 0 of 35 customers meet their OTIF target.

One signal, five metrics, drilled from months to days

Metric switcher

A **disconnected Metrics table** + **SELECTEDVALUE** router drives OT / IF / OTIF / LIFR / VOFR on the same trend chart

Drill path

Months



Weeks

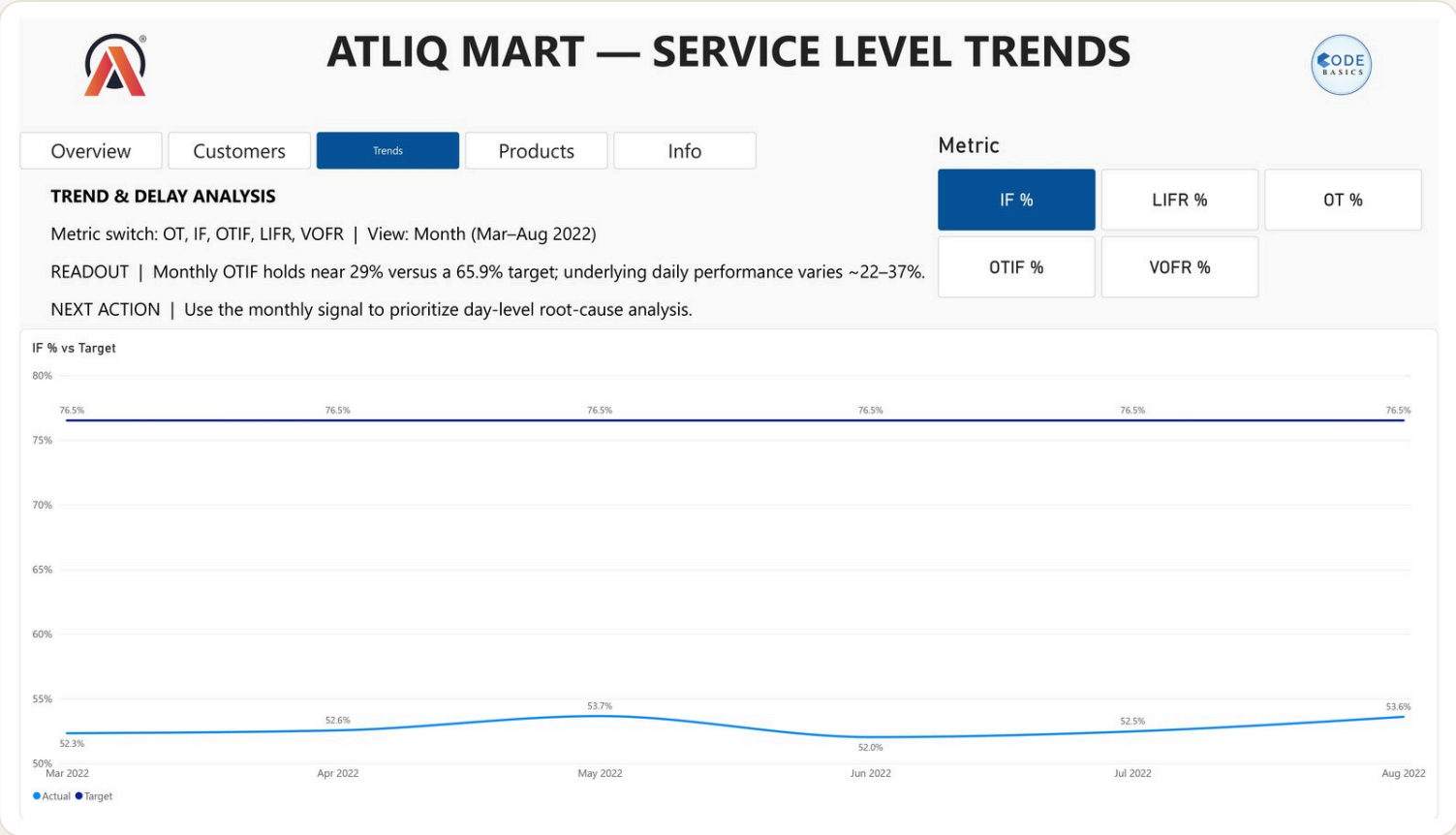


Days

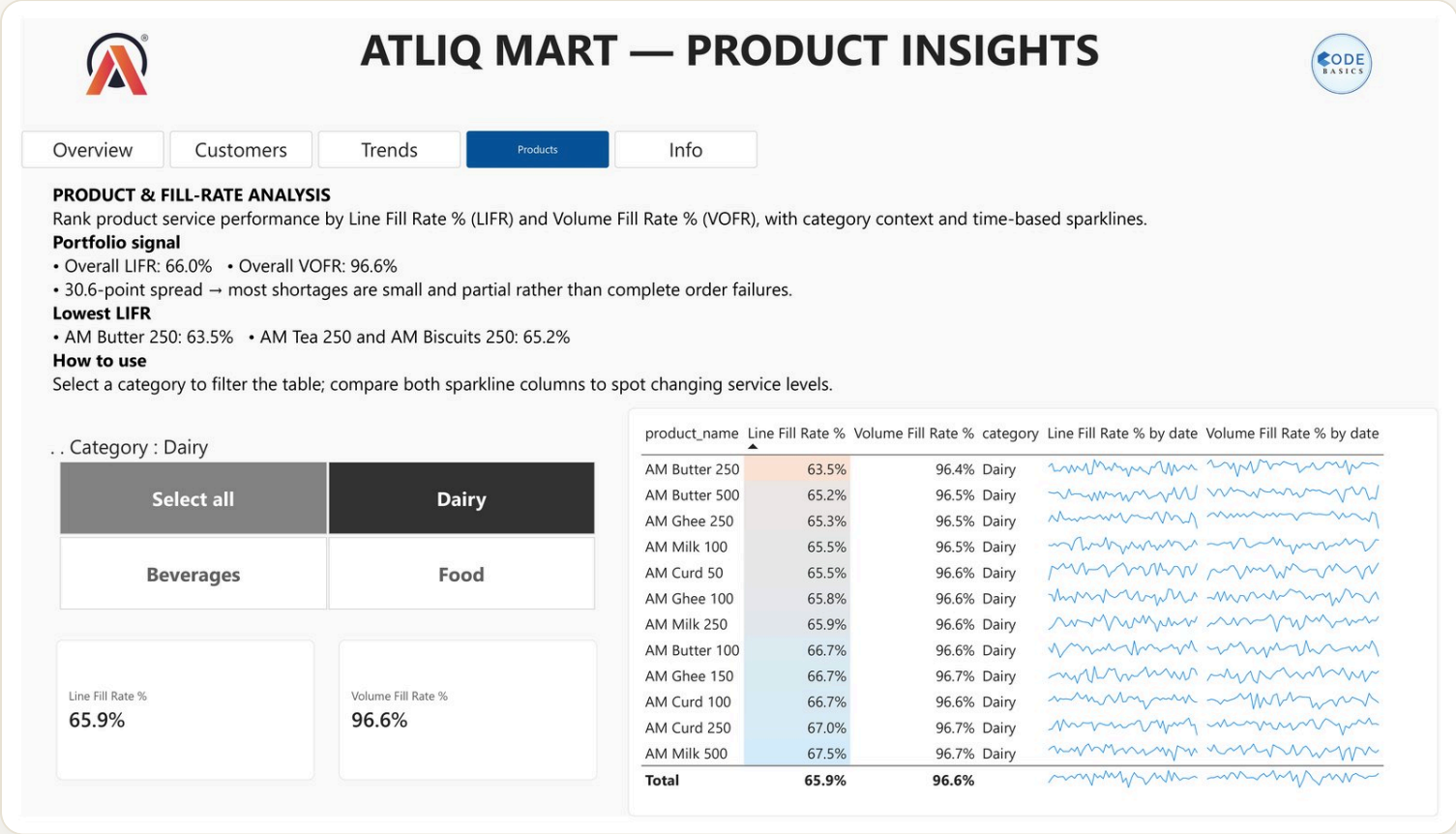
exposing how unstable daily performance really is — daily OTIF swings **22–37%**

Dynamic chart title

A DAX title measure shows the selected metric and current drill level, using **ISFILTERED**



Fill rates by product, with time embedded in every row



Under half the promised reliability — everywhere

29.0%

ACTUAL OTIF

65.9%

TARGET

↘ -36.9 pts below target



Both pillars are short

OT **59.0%** vs **86.1%** target (**-27.1 pts**) · IF **52.8%** vs **76.5%** target (**-23.7 pts**) — on-time and in-full fail together, so OTIF compounds the damage



Systemic, not regional

City OTIF sits in a narrow **27.8–30.1%** band — every city fails the same way, so no local hero story explains it



Flat but erratic

Monthly OTIF barely moves (**28.7–29.4%**) while daily swings reach **22–37%** — the average hides the volatility

! 0 of 35 customers meet their OTIF target

Two independent failure modes — not one problem

MODE A



the big-3 chains collapse on on-time

Coolblue

13.7%

Acclaimed

15.5%

Lotus

16.3%

~30% of all orders · on-time only ~29%

The largest accounts are late on most orders — everything downstream of dispatch fails for them



Shortages are small and partial

LIFR 66.0% yet VOFR 96.6% — orders are almost complete, not failed; fixing pick accuracy and SKU-level stock recovers most of the gap

MODE B



mid-tier accounts ship on time but under-fill

Info Stores

Elite Mart

Sorefoz

Vijay

on-time

~72%

in-full

only ~40%

~23% of orders

These accounts arrive when promised — the boxes are just light



Lateness is frequent but shallow

28.9% of lines ship late — average 1.7 days, never more than 3: a scheduling problem, not a transport failure

Two parallel workstreams, sequenced by account value



Workstream 1 — Dispatch & scheduling

attacks **Mode A**: the on-time collapse of the big-3 chains



Workstream 2 — Supply planning

attacks **Mode B**: the partial fills of the mid-tier accounts

01

Open renewal conversations with the big-3 accounts first

they are ~30% of order volume; retention starts where the risk concentrates



02

Add a 1–2 day scheduling buffer rather than redesigning transport

lateness is shallow (avg **1.7 days**, never more than **3**) — a calendar fix, not a fleet fix



03

Hold SKU-level safety stock and review pick accuracy

the shortfalls are small and partial (VOFR **96.6%**), so warehouse discipline recovers most of the in-full gap



 Pilot the fixes in Vadodara — the weakest and highest-volume city — before scaling to the expansion metros.

Verified numbers, open artifacts

 VERIFIED NUMBERS

59.0%	52.8%	29.0%	66.0%	96.6%
OT	IF	OTIF	LIFR	VOFR

31,729	57,096	35	18	6
ORDERS	LINES	CUSTOMERS	PRODUCTS	TABLES



Power Query (M)

cleaning & shaping



Power BI Desktop

star schema + DAX



Python / pandas

verification



Markdown docs

measures reference + project journal

 The repo ships the **.pbix dashboard**, the **6 source CSVs**, a full **PDF report**, an executive deck, and the **docs/ journal**.

 **ATLIQ MART — DEFINITIONS & DATA MODEL** 

OverviewCustomersTrendsProducts

Key Metrics & Definitions

On Time % (OT %) — % of orders delivered on or before the agreed date. Order-level.
In Full % (IF %) — % of orders where ALL lines shipped complete. Order-level.
OTIF % — % of orders BOTH on time AND in full. The reliability metric.
Line Fill Rate % — % of order lines shipped in full quantity. Line-level.
Volume Fill Rate % — Total qty shipped ÷ total qty ordered. Line-level.

Targets: per-customer targets from dim_targets.orders (OT, IF, OTIF only — LIFR and VOFR have no targets in the source data).

Gap = Actual – Target, in percentage points. Matrix colors: green ≥ target, yellow within –10, orange within –25, red beyond –25.

Data Model (Star Schema)
Fact tables: fact_order_lines (57k order lines), fact_orders_aggregate (32k orders)
Dimensions: dim_customers (35 accounts), dim_products (18 SKUs), dim_date (Mar–Aug 2022), dim_targets.orders (per-customer targets)
Relationships: All dimensions filter facts (single direction).
Date connects via order_placement_date.
fact-to-fact joins deliberately avoided (star schema discipline).
Data cleaning: Power Query — date parsing, header fixes (BOM), category capitalization, column renames, month_sort helper column.

Key Insights

— OTIF 29.0% vs 65.9% target: under half promised reliability across all 6 months
— LIFR 66.0% but VOFR 96.6%: shortages are small and partial (supply planning issue)
— Failure is uniform across cities (OTIF 27.8–30.1%): systemic, not regional
— Monthly results are flat but daily swings reach 22–37%: chronic and erratic
— Lowest OTIF accounts: Lotus Mart (16.3%), Acclaimed Stores (15.5%), Coolblue (13.7%)
— Lowest LIFR products: AM Butter 250 (63.5%), AM Tea 250 and AM Biscuits 250 (65.2%)
Notes
— Data window: 01-Mar-2022 to 31-Aug-2022
— Product view includes native LIFR and VOFR sparklines by date
— Metric switch and drill: trend chart supports months → weeks → days

📖 Page 5 documents the model so anyone can audit it

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Scenario & data: Codebasics Resume Project Challenge #2 — AtliQ Mart (codebasics.io). Dashboard design, analysis, and insights are the author's own work.